Structure-Constrained Behavior Generation: A Role-Semantic Invariant Framework for Behavioral Modes

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Abstract

We propose a theoretical framework for behavior generation in dynamical systems, termed Structure-Constrained Behavior Generation (SCBG). Unlike conventional approaches where behavior is derived from optimization or learned policies, SCBG posits that behavior emerges as a selection process constrained by structural invariants.

We introduce the concept of role-semantic structural invariants, extending traditional graph-based representations by incorporating functional roles associated with nodes and pathways. Within this framework, behavior is determined not by raw connectivity or parameter values, but by invariant structural relations including opposition, closure, hierarchical pathways, and role semantics.

Through a sequence of minimal experiments, we establish a consistent evidence chain showing that: (1) structural differences induce distinct behavioral modes, (2) perturbations of structural invariants lead to behavioral degradation, (3) structural influence dominates input mapping effects, and (4) behavior equivalence is preserved only under role-consistent structural transformations within a bounded scaling regime.

We formalize SCBG through a set of definitions, axioms, and theorems, demonstrating that behavioral modes are functions of role-embedded structural invariants rather than direct outputs of control policies.

This work suggests a shift in perspective from policy-driven behavior generation to structure-driven mode selection, providing a foundation for future studies in control systems, robotics, and dynamical behavior modeling.

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1. Introduction

Traditional behavior generation in dynamical systems follows a paradigm in which behavior is derived from: • explicit control laws (e.g., PID, MPC), • optimization processes, or • learned policies (e.g., reinforcement learning).

These approaches share a common structure:

$$\text{state} \rightarrow \text{policy / optimization} \rightarrow \text{action}$$

In contrast, this work explores an alternative viewpoint:

Behavior is not directly computed, but emerges as a constrained selection from a structure-defined set of reachable modes.

We term this paradigm Structure-Constrained Behavior Generation (SCBG).

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1. Core Formulation

The minimal SCBG formulation is:

$$S + I + U \Rightarrow B$$

Where: • S: structure • I: input • U: update rule • B: behavioral modes

Further, SCBG can be expressed as:

$$B = F(\mathrm{Inv}_r(S))$$

where $\mathrm{Inv}_r(S)$ denotes role-semantic structural invariants.

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1. Definitions

D1. Structure
Structure is defined as:
A set of allowed and forbidden state transition relations, augmented with role semantics.
A numerical matrix is only an implementation, not the essence of structure.

D2. Behavioral Modes
Behavioral modes are defined as:
Stable or metastable attractor classes under structural constraints.

D3. Generation
Generation is defined as:
A constraint-driven selection process from reachable states into a behavioral mode.

D4. Behavioral Equivalence
Two structures S_1 and S_2 are equivalent if:
$S_1 \sim S_2 \iff B(S_1) \approx B(S_2)$

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1. Structural Invariants

SCBG identifies four core invariants:

I1. Opposition
Opposing functional relations (e.g., approach vs avoidance) are essential. Violation leads to behavioral collapse.

I2. Closure
Closed structural loops ensure: • stability • recoverability • consistent behavior

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I3. Hierarchical Pathways
Distinct functional roles must exist: • driving pathways • modulating pathways

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I4. Role-Semantic Preservation
Nodes are not anonymous; they carry functional meaning. Valid transformations must preserve: • node roles • input-role mapping

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1. Axioms

A1. Structure Dominance
Behavioral modes are primarily determined by structural invariants.

A2. Input Modulation
Inputs select among available modes but do not define them.

A3. Finite Mode Set
For fixed structure and bounded input, only finite behavioral modes exist.

A4. Perturbation Hierarchy
$\Delta(W) \gg \Delta(G)$
Structural perturbations dominate input perturbations.

A5. Valid Mode Constraint
Only behaviors satisfying structural invariants are valid.

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1. Theorems

T1. Structural Equivalence Theorem
If two structures preserve: • invariants • role semantics • bounded parameter scaling then:
$S_1 \sim S_2$

T2. Structural Breakdown Theorem
Violation of invariants leads to: • oscillatory behavior • locked states • failure modes

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T3. Input Modulation Theorem
With fixed structure:
$\mathcal{B}(S, I_1) = \mathcal{B}(S, I_2)$

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1. Evidence Chain

- Structural variation → behavioral variation
- Structural perturbation → degradation
- Structure dominates input
- Role-preserving transformations maintain equivalence
- Scaling invariance holds within bounded window

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1. Core Statement

Behavior is determined by role-semantic structural invariants, while inputs only select among predefined modes.

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1. Scope and Limitations

Valid scope: • small-scale structured systems • systems with clear role assignments
Limitations: • no full mathematical proof • no large-scale validation • no real-world robotics deployment

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1. Conclusion

SCBG introduces a new paradigm:
Structure-driven mode selection instead of policy-driven behavior generation.